LM Physics of Climate

Anjali A. A. Piette

8 Radiative equilibrium for a grey gas

8.1 The grey gas model

As we saw in Section 5, atmospheric absorption is highly dependent on frequency. This makes it challenging to solve the radiative transfer and radiative equilibrium equations, and this is typically done numerically in atmospheric simulations. However, we can still gain important insights by assuming that the gas absorbs uniformly across all frequencies, i.e. κ_ν is independent of ν, becoming the constant value κ (similarly, σ_ν becomes constant with frequency and can be written as σ). This is referred to as the *grey gas approximation*.

So far, we have mostly been using the two-stream equations to describe fluxes per unit frequency ($\mathcal{F}_+$ in the upwards direction, and $\mathcal{F}_-$ in the downwards direction). Assuming that κ_ν is independent of frequency makes it much easier to integrate these equations over frequency. We will then be dealing with fluxes that have been integrated over wavelength (i.e., energy per unit time per unit area, and *not* per unit frequency). We will denote these fluxes as

$$F_+ = \int_0^\infty \mathcal{F}_+ \mathrm{d}\nu \tag{8.1}$$

$$F_- = \int_0^\infty \mathcal{F}_- \mathrm{d}\nu \tag{8.2}$$

We can also integrate both two-stream equations (Equations 6.9 and 6.10) over all frequencies. In the upwards direction,

$$\begin{aligned} \int_0^\infty \frac{\mathrm{d}\mathcal{F}_+}{\mathrm{d}\overline{\tau}} \mathrm{d}\nu &= \int_0^\infty (-\mathcal{F}_+ + \pi B_\nu(T)) \mathrm{d}\nu \\ \frac{\mathrm{d}F_+}{\mathrm{d}\overline{\tau}} &= -F_+ + \sigma_{SB} T(\overline{\tau})^4, \end{aligned} \tag{8.3}$$

and in the downwards direction,

$$\begin{aligned} \int_0^\infty \frac{\mathrm{d}\mathcal{F}_-}{\mathrm{d}\overline{\tau}} \mathrm{d}\nu &= \int_0^\infty (\mathcal{F}_- - \pi B_\nu(T)) \mathrm{d}\nu \\ \frac{\mathrm{d}F_-}{\mathrm{d}\overline{\tau}} &= F_- - \sigma_{SB} T(\overline{\tau})^4, \end{aligned} \tag{8.4}$$

where we have used the fact that $\int_0^\infty \pi B_\nu(T) \mathrm{d}\nu = \sigma_{SB} T^4$.

8.2 Radiative equilibrium for a grey atmosphere

Now, we will calculate the temperature profile of a grey atmosphere in radiative equilibrium. In radiative equilibrium, the heating rate is equal to zero at every altitude (and

hence every value of $\overline{\tau}$) in the atmosphere. Recall that the heating rate per unit optical thickness is $h_\nu = -\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(\mathcal{F}_+ - \mathcal{F}_-)$ (Equation 7.7). In the grey gas approximation, we can integrate this over all frequencies to get

$$h = \int_0^\infty h_\nu \mathrm{d}\nu = -\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-). \tag{8.5}$$

So, when $h = 0$, $F_+ - F_-$ is independent of $\overline{\tau}$ and takes a constant value. We can work out the value of this constant by considering the boundary condition at the top of the atmosphere. Here, we will assume that there is no incoming downward flux, so $F_-(\overline{\tau}_\infty) = 0$. The outgoing flux, $F_+(\overline{\tau}_\infty)$ is the total emission of the planet out to space. It is commonly labelled the *Outgoing Longwave Radiation*, or OLR. Why longwave? Because here we are considering the thermal emission from the planet, which is dominated by infrared radiation (in contrast, optical radiation from the Sun is referred to as 'shortwave'). So, the integral across all frequencies is actually dominated by the infrared contribution.

Note that while we have set $F_-(\overline{\tau}_\infty) = 0$, we aren't completely ignoring radiation from the Sun. We are effectively assuming that the atmosphere is transparent to the Sun's radiation, so that it heats up the surface directly. That way, heating from the Sun is taken into account by the thermal emission from the surface.

The value of the OLR is set by energy balance: in equilibrium, the energy received from the Sun perfectly balances the total thermal emission from the planet (which is the OLR). This is the calculation we did back in Section 1 to work out the equilibrium temperature of the planet. In that calculation, we treated the planet as a blackbody radiator and worked out what temperature, T_{eq}, it would need to be at such that the planet's thermal emission would balance the incoming stellar radiation. Thus, $OLR = \sigma_{SB} T_{eq}^4$.

We can rewrite Equation 8.5 by taking the difference of Equations 8.3 and 8.4. This allows us to link radiative equilibrium to the temperature profile of the atmosphere. In radiative equilibrium, we have:

$$\begin{aligned} 0 = \frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ - F_-) &= (-F_+ + \sigma_{SB} T(\overline{\tau})^4) - (F_- - \sigma_{SB} T(\overline{\tau})^4) \\ &= -(F_+ + F_-) + 2\sigma_{SB} T(\overline{\tau})^4. \end{aligned} \tag{8.6}$$

We can also take the sum of Equations 8.3 and 8.4, giving:

$$\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ + F_-) = -(F_+ - F_-). \tag{8.7}$$

We can solve this differential equation by using that fact that $F_+ - F_-$ is a constant (since its derivative is equal to zero from Equation 8.5). As noted above, $F_+ - F_- = OLR$, so $\frac{\mathrm{d}}{\mathrm{d}\overline{\tau}}(F_+ + F_-) = -OLR$. Before we integrate this equation, let's choose some helpful boundary conditions: at $\overline{\tau} = \overline{\tau}_\infty$, we know that $F_+ + F_- = OLR$. Integrating Equation 8.7 then gives:

$$\begin{aligned} \int_{F_+ + F_-}^{OLR} \mathrm{d}(F_+ + F_-)' &= -OLR \int_{\overline{\tau}}^{\overline{\tau}_\infty} \mathrm{d}\overline{\tau}' \\ OLR - (F_+ + F_-) &= -(\overline{\tau}_\infty - \overline{\tau}) OLR \\ F_+ + F_- &= (1 + \overline{\tau}_\infty - \overline{\tau}) OLR. \end{aligned} \tag{8.8}$$

We can now substitute this back into Equation 8.6:

$$\begin{aligned} 2\sigma_{SB}T(\overline{\tau})^4 &= (F_+ + F_-) = (1 + \overline{\tau}_\infty - \overline{\tau})OLR \\ T &= \left(\frac{(1 + \overline{\tau}_\infty - \overline{\tau})OLR}{2\sigma_{SB}}\right)^{1/4} \\ &= \left(\frac{(1 + \overline{\tau}_\infty - \overline{\tau})}{2}\right)^{1/4} T_{eq}. \end{aligned} \tag{8.9}$$

This is the temperature profile of a grey atmosphere in radiative equilibrium which is transparent to incoming optical radiation. It's worth noting a few behaviours of this temperature profile. Firstly, note that at the upper edge of the atmosphere, where $\overline{\tau} = \overline{\tau}_\infty$, the temperature becomes $T_{eq}/2^{1/4}$, which is the *skin temperature* we derived in Equation 1.3. If the atmosphere is optically thin, $(\overline{\tau}_\infty - \overline{\tau}) \ll 1$ throughout the atmosphere, and the temperature is isothermal at the skin temperature. If the atmosphere is optically thick, the temperature gently decreases from the surface until it reaches the skin temperature.

8.3 Radiative-convective equilibrium for a grey atmosphere

The temperature profile in Equation 8.9 can become very steep near the surface if $\overline{\tau}_\infty$ is large. If it becomes steep enough, the atmosphere will become unstable to convection according to the Schwarzschild criterion (Equation 2.28) for dry gas, the Clausius-Clapeyron equation (Equation 3.8) for a single-component condensing atmosphere, or the two-component moist pseudo-adiabat (Equation 4.6) for a mixture of a condensable and non-condensable gas.

So, which parts of the atmosphere will become unstable to convection? We can work this out by converting Equation 8.9 into a '$\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P}$' gradient, and comparing with the criteria for convective instability. Let's start by taking the derivative of Equation 8.9 with respect to $\overline{\tau}$:

$$8\sigma_{SB}T^3\frac{\mathrm{d}T}{\mathrm{d}\overline{\tau}} = -OLR. \tag{8.10}$$

Then, using the chain rule, we can write:

$$\begin{aligned} 8\sigma_{SB}T^3\frac{\mathrm{d}T}{\mathrm{d}P}\frac{\mathrm{d}P}{\mathrm{d}\overline{\tau}} &= -OLR \\ \frac{8\sigma_{SB}T^4}{P}\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P}\frac{\mathrm{d}P}{\mathrm{d}\overline{\tau}} &= -OLR \\ \frac{\mathrm{d}\ln T}{\mathrm{d}\ln P} &= -\frac{\mathrm{d}\overline{\tau}}{\mathrm{d}P}\frac{P}{8\sigma_{SB}T^4}OLR \\ \frac{\mathrm{d}\ln T}{\mathrm{d}\ln P} &= -\frac{1}{4}\frac{\mathrm{d}\overline{\tau}}{\mathrm{d}P}\frac{P}{1 + \overline{\tau}_\infty - \overline{\tau}}, \end{aligned} \tag{8.11}$$

where in the last step we have substituted Equation 8.9.

Recall that we can convert between $\mathrm{d}\overline{\tau}$ and $\mathrm{d}P$ by using the definition of optical depth and the hydrostatic equilibrium equation: $\mathrm{d}\tau = \mathrm{d}\overline{\tau}\ \overline{\mu_\theta} = \kappa\rho\mathrm{d}z = -(\kappa/g)\mathrm{d}P$. Integrating this gives:

$$\begin{aligned} \int_{\tau_\infty}^{\tau}\mathrm{d}\tau' &= -\frac{\kappa}{g}\int_0^P \mathrm{d}P' \\ \tau - \tau_\infty &= -\frac{\kappa}{g}P. \end{aligned} \tag{8.12}$$

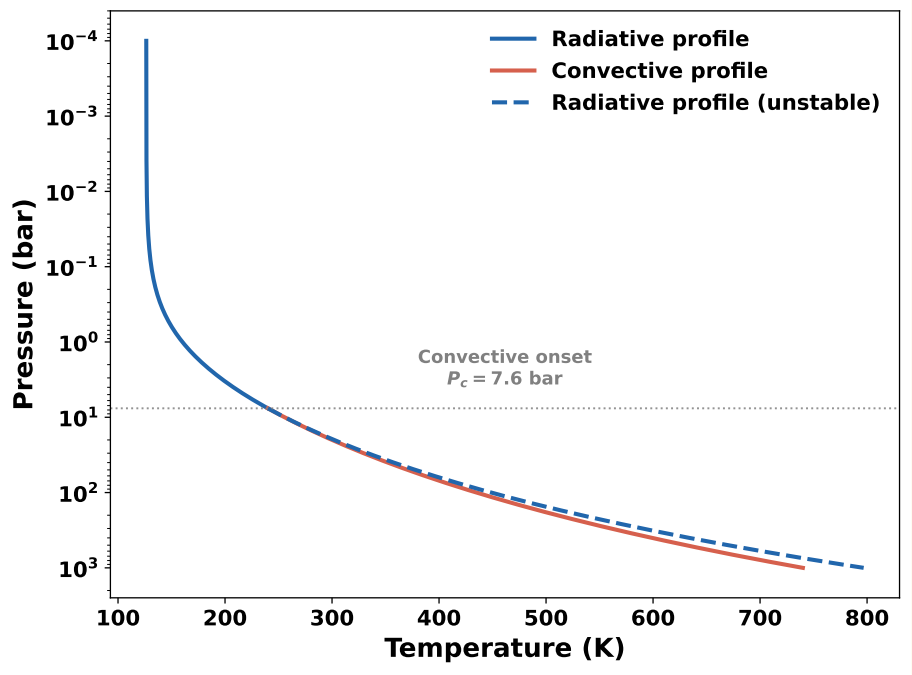


Figure 1: The temperature profile of an atmosphere in radiative-convective equilibrium. In the upper atmosphere, the temperature profile follows Equation 8.9. In the deep atmosphere (below the dashed grey line), the radiative temperature profile (dashed blue line) is unstable to convection, so the atmosphere follows an adiabatic temperature gradient (in this example, a dry adiabat is plotted in the solid red line).

Thus, noting that $\tau - \tau_\infty = \overline{\mu_\theta}(\overline{\tau} - \overline{\tau}_\infty)$, the factor of $\frac{\mathrm{d}\overline{\tau}}{\mathrm{d}P}P$ in Equation 8.11 simplifies to $-\frac{\kappa P}{g\overline{\mu_\theta}} = \overline{\tau} - \overline{\tau}_\infty$. Substituting this back into Equation 8.11, we get

$$\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P} = \frac{1}{4}\frac{\overline{\tau}_\infty - \overline{\tau}}{1 + \overline{\tau}_\infty - \overline{\tau}}. \tag{8.13}$$

The maximum gradient occurs when $\overline{\tau} = 0$, where $\frac{\mathrm{d}\ln T}{\mathrm{d}\ln P} = \frac{1}{4}\frac{\overline{\tau}_\infty}{1+\overline{\tau}_\infty}$. This gradient is larger for larger values of $\overline{\tau}_\infty$, meaning that more optically thick atmospheres have steeper temperature profiles close to their surface, and are less stable against convection. For a dry atmosphere, the condition for convection to occur is (following Equation 2.28):

$$\frac{1}{4}\frac{\overline{\tau}_\infty - \overline{\tau}}{1 + \overline{\tau}_\infty - \overline{\tau}} \geq \frac{\gamma - 1}{\gamma}. \tag{8.14}$$

We can see how this plays out in an atmosphere in Figure 1. In the upper atmosphere, $\overline{\tau}_\infty - \overline{\tau}$ is small, so the left hand side of Equation 8.14 is small, and the atmosphere is stable against convection. Where $\frac{1}{4}\frac{\overline{\tau}_\infty - \overline{\tau}}{1+\overline{\tau}_\infty - \overline{\tau}} = \frac{\gamma-1}{\gamma}$, the atmosphere becomes unstable against convection (this happens at a pressure of 7.6 bar in the example in Figure 8.14). Deeper than this, the atmosphere is convective, and follows an adiabatic gradient (the example in Figure 8.14 is for a dry adiabat).